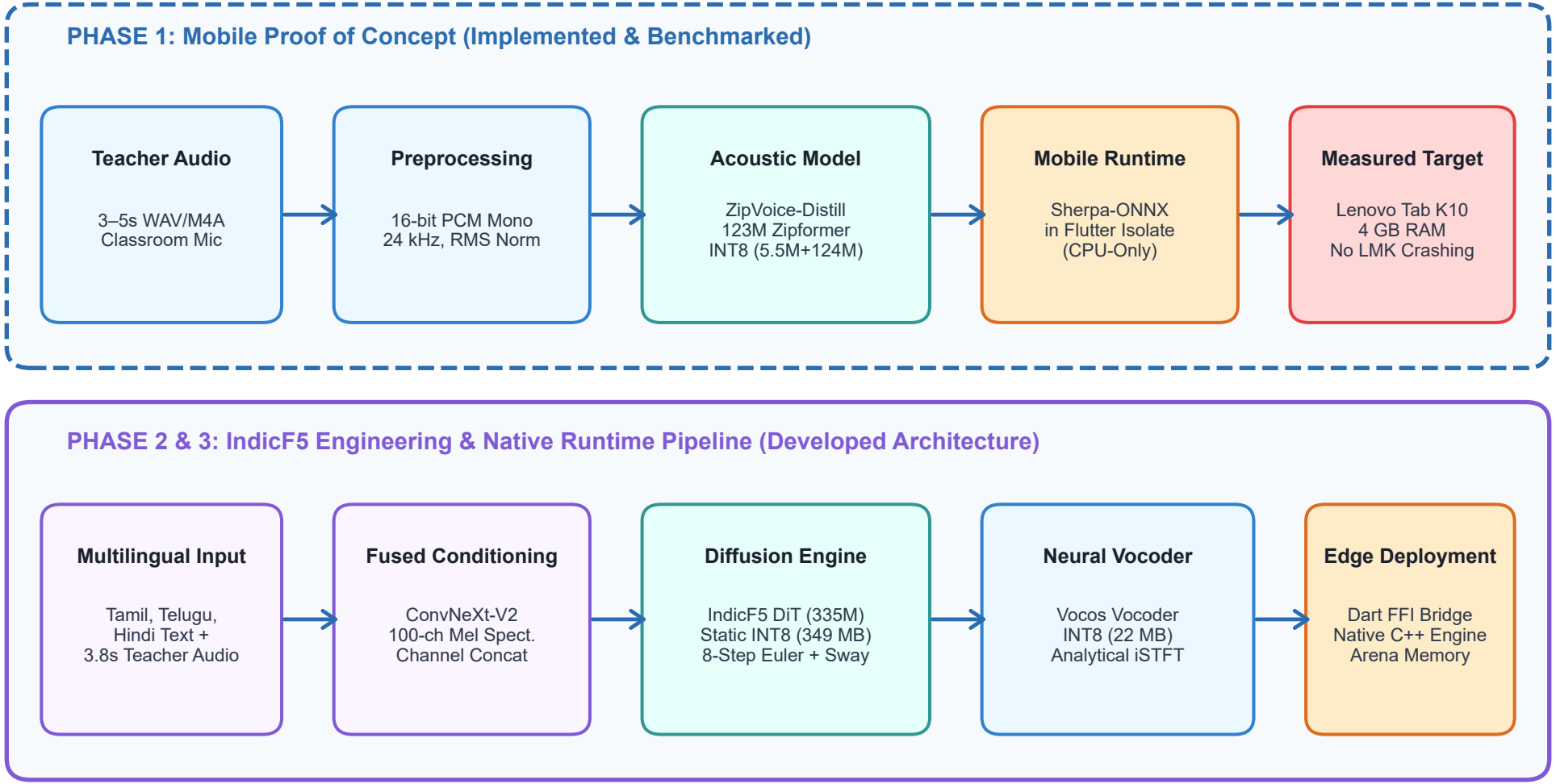

PHASE 1: Mobile Proof of Concept (Implemented & Benchmarked)
Teacher Audio
3–5s WAV/M4A
Classroom Mic
Preprocessing
16-bit PCM Mono
24 kHz, RMS Norm
Acoustic Model
ZipVoice-Distill
123M Zipformer
INT8 (5.5M+124M)
Mobile Runtime
Sherpa-ONNX
in Flutter Isolate
(CPU-Only)
Measured Target
Lenovo Tab K10
4 GB RAM
No LMK Crashing
PHASE 2 & 3: IndicF5 Engineering & Native Runtime Pipeline (Developed Architecture)
Multilingual Input
Tamil, Telugu,
Hindi Text +
3.8s Teacher Audio
Fused Conditioning
ConvNeXt-V2
100-ch Mel Spect.
Channel Concat
Diffusion Engine
IndicF5 DiT (335M)
Static INT8 (349 MB)
8-Step Euler + Sway
Neural Vocoder
Vocos Vocoder
INT8 (22 MB)
Analytical iSTFT
Edge Deployment
Dart FFI Bridge
Native C++ Engine
Arena Memory